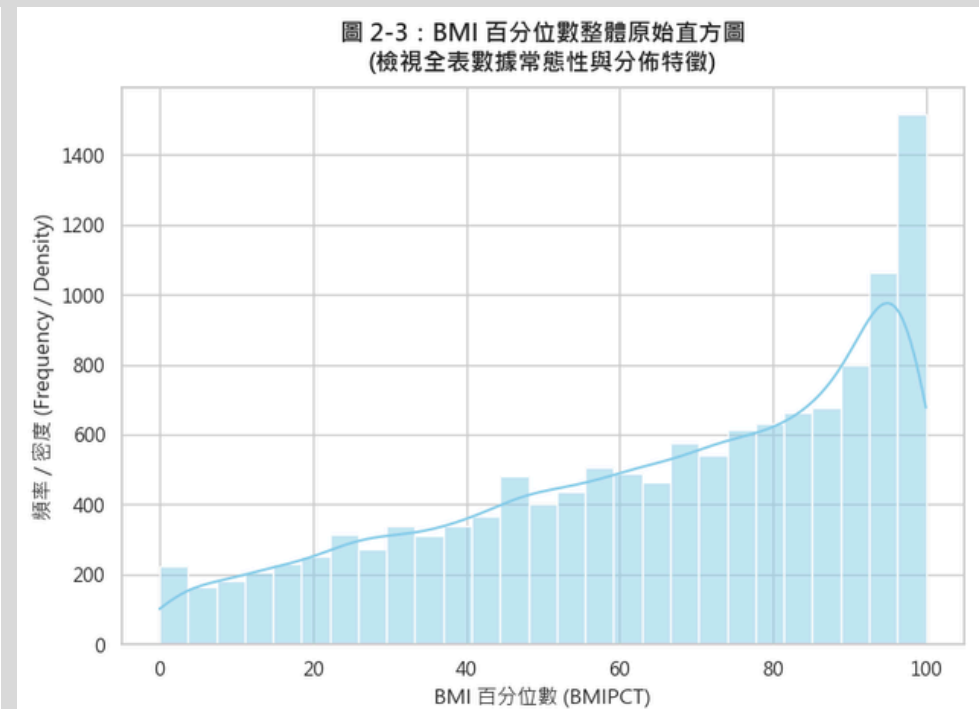
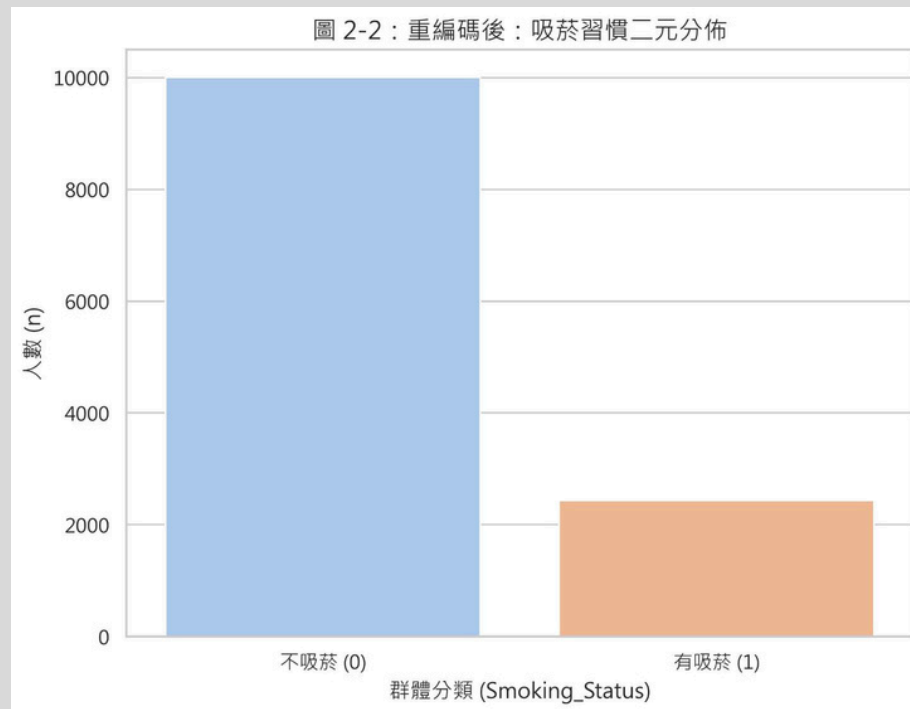


Group 14 Adolescent Behavioral Metrics & Health Profiles

1 Research Question & Data

- **Question:** Is there a significant difference in the mean BMI percentile between students who currently use cigarettes and those who do not?
- **Dataset:** YRBS 2007
- **Variables:**
 - **Group Variable:** CurrentCigaretteUse (Binary: 1 = Yes, 0 = No)
 - **Response Variable:** BMIPCT (Continuous: BMI Percentile)

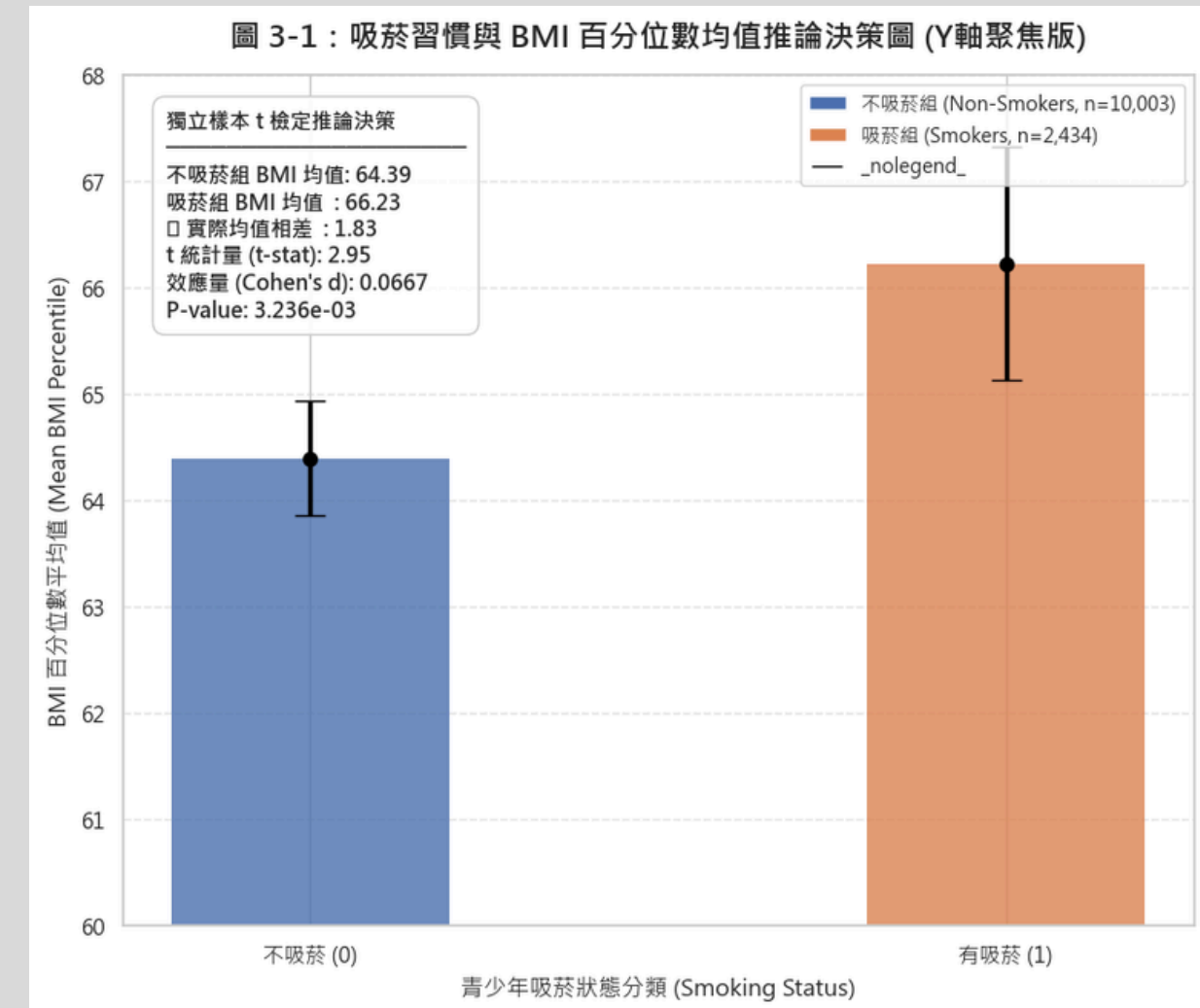
2 EDA Highlights



- **Group Variable Imbalance:** Figure 2-2 highlights a difference in sample sizes between the two groups. Non-smokers (Group 0, ~10,000 students) significantly outnumber smokers (Group 1, ~2,400 students). However, both groups are sufficiently large for a robust two-sample t-test.
- **Response Variable (BMIPCT) is Left-Skewed:** Figure 2-3 shows that the overall BMI percentile distribution is visibly left-skewed, with a large concentration of students in the higher percentiles.

3 Statistical Inference

- ✓ **Decision:** Reject H_0 ($p < 0.05$). There is a statistically significant difference between the two means.



4 Conclusion

1. **Significant Difference:** Statistical results indicate that students with current cigarette use have a significantly higher mean BMI percentile (about 1.83 percentiles higher) than non-smokers.
2. **Observational Limitation (No Causation):** Since this is an observational study, the positive association does not imply that smoking causes an increase in BMI.
3. **Potential Confounders:** This difference may be driven by other underlying lifestyle factors (e.g., dietary habits, physical activity) or psychological variables.